Venkata Lakshmi Parimala Pasupuleti

Deployed Engineer · AI Agents · Production Software Engineering · Python · TypeScript · Customer Deployment & Adoption · Claude Code & Cursor Daily User

venkatalakshmiparimala@gmail.com | +1-703-261-5410 | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Senior Software Engineer who has both *built* AI software agents and *uses* AI coding agents as core workflow tools every day — the combination that lets a Deployed Engineer work as a genuine peer with any customer engineering team adopting Devin. Leasel is a production 8-agent LangGraph system (planning, tool-use, durable state, HITL escalation); AgentProbe is a homegrown evaluation framework that measures exactly whether AI agents are reliably doing what they're supposed to — the first question every customer asks. 8 years at Entrata owning production debugging end-to-end: 20+ modules, 99.5% uptime, self-directed incident resolution with no PM. Daily user of Claude Code and Cursor. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and AI-integrated pipelines; working directly with the founding team to translate technical challenges into solutions; using AI coding agents (Claude Code, Cursor) as core development tools throughout; debugging production reliability issues and driving feature adoption.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern · Remote

- Built and deployed a full agentic AI system from scratch (LangGraph): orchestration, tool use, RAG over regulatory corpus, guardrails, and conditional routing — operated beyond job description to own customer deployment and integration end-to-end. Identified high-impact use cases independently and drove implementation without spec.

George Mason University
Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Delivered technical workshops to 600+ students (Python, SQL, OOP, automation) — translated complex engineering concepts clearly to any technical level; provided one-on-one debugging support across diverse problems. Research: built multi-source ML pipelines (Python, R) across 190+ countries ($R^2 = 0.977$); HuggingFace fine-tuning + PyTorch deployment.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Self-directed production ownership under ambiguity for 7 years** (Python, Java, PHP, PostgreSQL, AWS, Kubernetes) — owned 20+ production modules at 99.5% uptime for 100K+ MAU with no PM or spec. Operated as the de-facto owner of production incident resolution: diagnosed, triaged, and fixed root causes across unfamiliar systems, rapidly learning codebases no one had documented. Built internal debugging tools that reduced incident resolution time by 40% and kept bugs below 10 for 6+ consecutive months.
- **Technical communication + customer deployment** — translated operational pain points into engineering solutions for cross-functional stakeholders; led AWS S3 document migration integrating Affordable Housing module (100% PHI compliance, zero incidents); mentored 5+ engineers; drove technical onboarding across 300+ code reviews; collaborated with global engineering teams in India across 120+ Agile sprints; eliminated 80% of manual workload through Python/Bash automation pipelines.

PROJECTS

AgentProbe: Does the AI Agent Actually Work? - Evaluation & Reliability Framework
TASK COMPLETION · CI REGRESSION

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- Built to answer the first question every customer asks after deploying an AI agent: "Is it actually working?" Measures task completion rate, time-to-resolution, escalation rate, hallucination rate, and groundedness; LLM-as-judge classifiers; CI regression harness with golden datasets so agent behavior can't silently degrade; multi-provider LLM routing with cost/latency controls; trace-level observability for debugging why an agent failed. Used to evaluate Leasel across 1,240+ compliance test cases (93.6% precision, 87.9% F1).

Leasel: Production 8-Agent AI System - Built, Deployed, Evaluated
LANGGRAPH · HITL + GUARDRAILS

Python · LangGraph · FastAPI · ChromaDB · MCP · SQLite · OpenAI API · LangSmith | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production agentic AI system: 8-agent LangGraph pipeline (Triage → Retrieval → Configuration → Historical → Reasoning → Validation → Escalation) with tool-use orchestration, durable state, contextual memory, error recovery, and HITL escalation when confidence falls below threshold. Designed, built, deployed, and evaluated end-to-end — the same lifecycle a Deployed Engineer owns for every Devin customer.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

AI Agents & Evaluation: LangGraph, LangChain, LangSmith, MCP, Autonomous AI Agents, Agentic AI, Multi-Agent Orchestration, Tool Calling, Durable State Management, HITL Escalation, Safety Guardrails, Error Recovery, Agent Reliability, Task Completion Metrics, CI Regression Harnesses, LLM-as-Judge, Hallucination Detection, Trace-Level Observability, RAG Pipelines, Embeddings, Hybrid Search, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, ChromaDB, Guardrails, Prompt Injection Defense, OWASP LLM Top 10, Red-Teaming, AI Coding Agents (Claude Code, Cursor)

Engineering & Deployment: Python (8+ yrs), TypeScript, JavaScript, Java (Spring Boot), Kotlin, SQL, Bash — Production Debugging, Customer Deployment, Product Integration, Technical Onboarding, Customer Adoption, Solution Deployment, FastAPI, REST APIs, PostgreSQL, MySQL, Redis, RabbitMQ, Docker, Kubernetes (K8s), AWS (S3, EC2, RDS, Lambda, Bedrock), CI/CD (GitHub Actions), Terraform, OpenTelemetry, Prometheus, Grafana, Distributed Systems, Microservices, System Design, Multi-Tenant Architecture, PHI Compliance, HIPAA

Communication & Execution: Technical Workshops (600+ students), Cross-Functional Stakeholder Communication, Code Reviews (300+), Global Distributed Teams, Agile/Scrum (120+ sprints), Self-Directed Ownership, Production Incident Resolution, Technical Writing, Mentoring (5+ engineers)